



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and spending patterns.

Dataset at a Glance

3,900

Transactions

18

Features

\$59.76

Avg. Purchase

3.75

Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings,
Shipping



Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing Review Ratings using category median

03

Feature Engineering

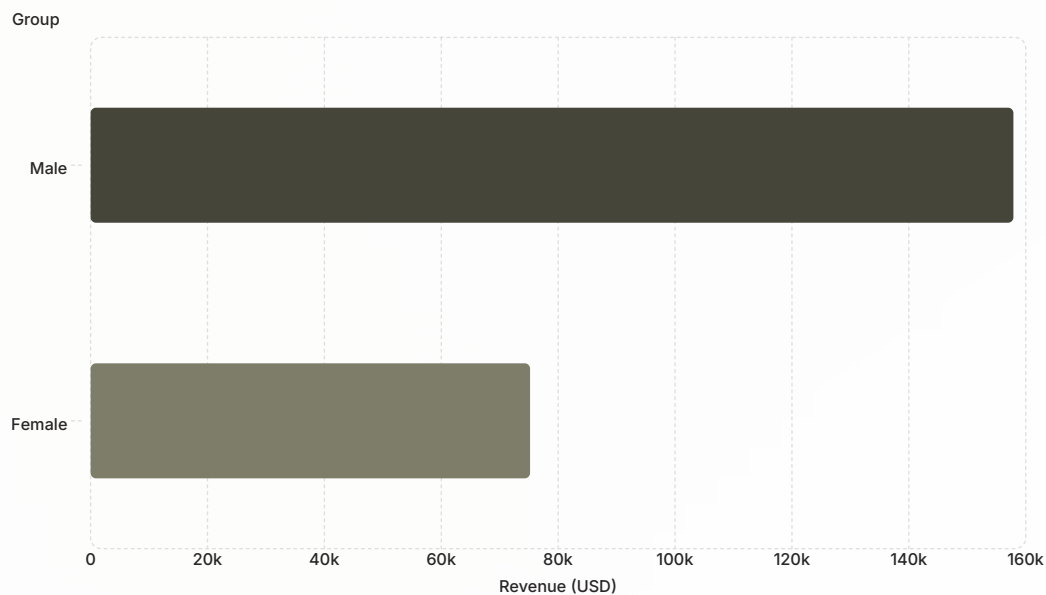
Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Connected Python to PostgreSQL; dropped redundant `promo_code_used`

Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- **839 discount users** still spent above average
- Express shipping avg: **\$60.48** vs. Standard: **\$58.46**

Top-Rated Products & Discount Leaders

Top 5 by Rating

 Gloves

3.86 ★

 Sandals

3.84 ★

 Boots

3.82 ★

Hat

3.80 ★

Skirt

3.78 ★

Most Discount-Dependent

Hat

50.00%

Sneakers

49.66%

Coat

49.07%

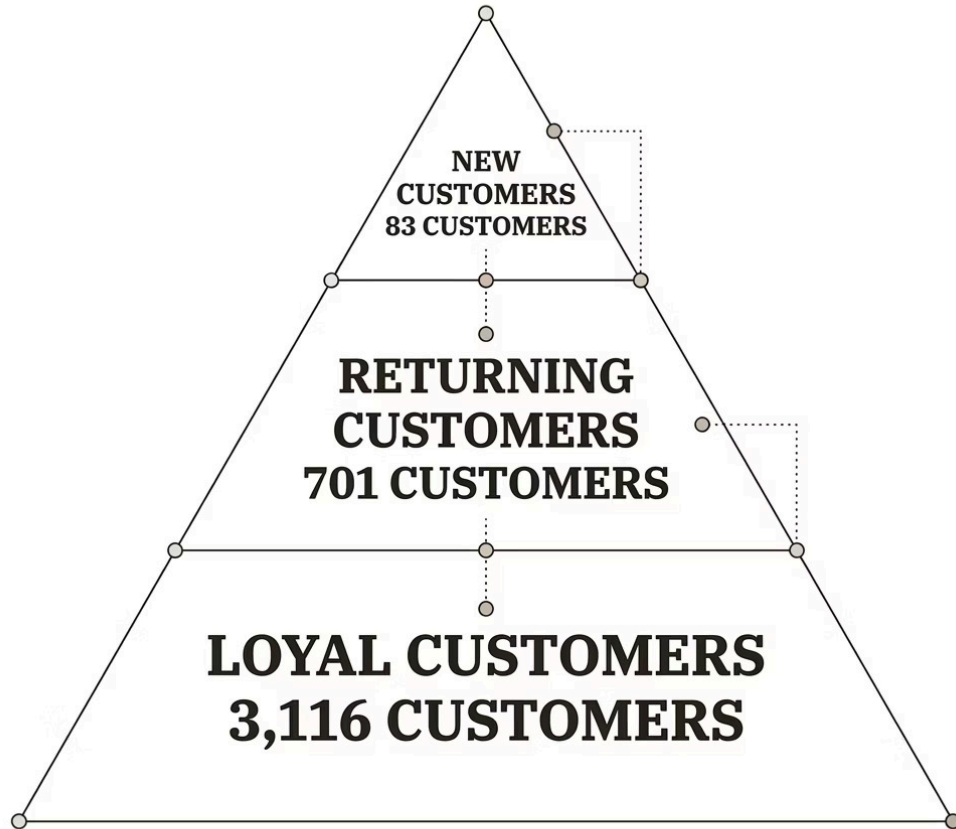
Sweater

48.17%

Pants

47.37%

Customer Segmentation



Segment Breakdown

Loyal

3,116 customers

Returning

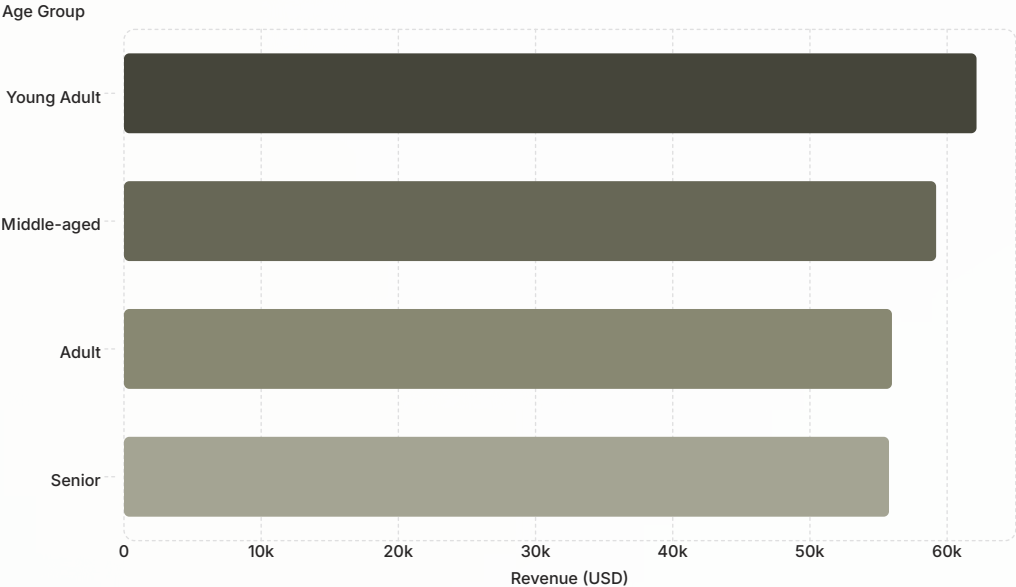
701 customers

New

83 customers

Customers with >5 purchases: 958 subscribed vs. 2,518 non-subscribed.

Subscriptions & Revenue by Age



Subscribers vs. Non-Subscribers

Status	Avg Spend	Total Rev.
Yes (1,053)	\$59.49	\$62,645
No (2,847)	\$59.87	\$170,436

Young Adults drive the highest revenue at **\$62,143**.

Top Products by Category

Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

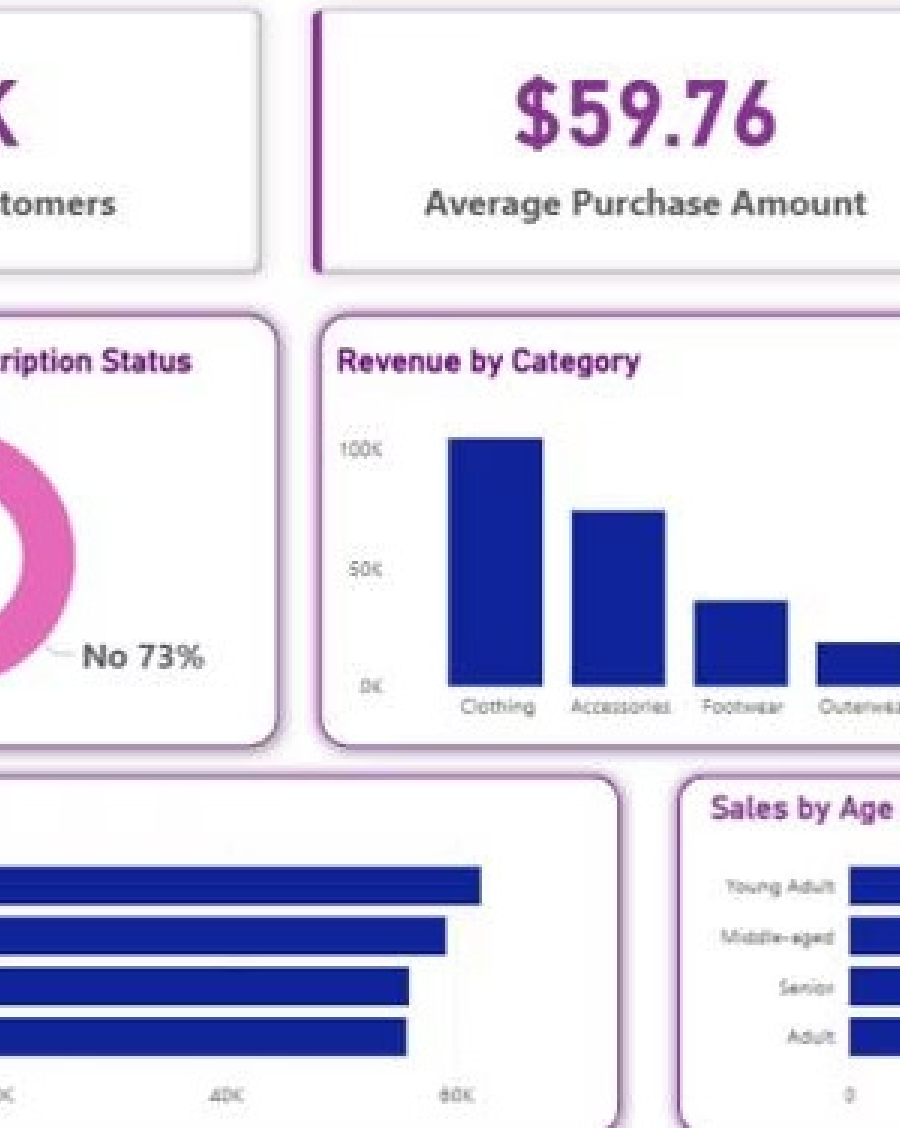
Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

Outerwear

- Jacket — 163
- Coat — 161

Customer Behavior Dashboard



Power BI Dashboard

3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

27% / 73%

Subscriber Split



Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits

→ Loyalty Programs

Reward repeat buyers to grow the Loyal segment

→ Review Discount Policy

Balance sales boosts with margin control

→ Targeted Marketing

Focus on high-revenue age groups & express-shipping users